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Evaluating Large Language Models in Engineering Education: A Multi-Course Assessment Across Bloom's Taxonomy

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ABSTRACT Large language models (LLMs) such as GPT-5, Claude, and Grok are increasingly incorporated into engineering education, yet their effectiveness on discipline-specific cognitive tasks remains insufficiently characterized. Engineering learning relies heavily on conceptual understanding, multi-step problem solving, and higher-order reasoning, making it essential to evaluate how LLMs perform across these contexts. This study investigates the performance of three LLMs across four undergraduate engineering courses, *Thermodynamics*, *Computer Programming*, *Systems and Control*, and *Circuits*, using multiple-choice questions aligned with Bloom's Taxonomy. We examine differences in accuracy and reasoning quality across cognitive levels, compare zero-shot and context-assisted prompting, and benchmark LLM performance against human learners in *Computer Programming*. Forty-eight instructor-designed questions (12 per course) spanning all Bloom levels were administered to each model. Responses were evaluated using accuracy, a rubric-based reasoning score, and a Composite Performance Index capturing completeness, efficiency, and quality of support. Human performance data were collected through an IRB-approved assessment with undergraduate students. Results show that Claude demonstrated the strongest overall performance and consistency, GPT-5 excelled in analytical reasoning, and Grok exhibited variable accuracy with limited higher-order reasoning. Providing contextual information improved GPT-5's performance but had minimal impact on Claude or Grok. In Programming, Claude and GPT-5 outperformed human participants across all Bloom levels. These findings suggest that LLMs can support engineering learning when paired with instructional strategies emphasizing verification and critical evaluation. However, persistent limitations at higher cognitive levels highlight the importance of guided, responsible integration, supported by a practical instructional guide for effective LLM use across engineering tasks.

INDEX TERMS Large Language Models (LLMs), Engineering Education, Bloom's Taxonomy, AI-Assisted Learning, Performance Evaluation

I. INTRODUCTION

THE emergence of large language models (LLMs) such as GPT-5, Claude, and Grok has transformed the landscape of education by introducing new possibilities for content generation, tutoring support, and automated reasoning [1]. While these models have demonstrated strong performance on general knowledge assessments, their effectiveness in domain-specific, technical fields such as engineering remains less thoroughly understood [2]. Engineering education demands not only factual recall but also conceptual understanding, applied problem-solving, and the

interpretation of quantitative data, skills that may challenge even the most advanced AI systems.

Understanding how AI models perform under these conditions not only informs their current utility as educational tools but also sheds light on their limitations as reasoning agents in engineering contexts. The findings can help guide both instructional design and responsible integration of AI into engineering curricula while minimizing the challenges associated with AI in education [3], [4].

While LLMs offer powerful assistance in generating explanations and solving problems, their increasing acces-

sibility raises critical pedagogical concerns [5]. Foremost among these is the risk that students might overly rely on AI-generated answers, potentially diminishing their engagement with the underlying concepts and critical thinking processes [6]. Engineering education, in particular, emphasizes the development of analytical skills, problem-solving creativity, and the ability to synthesize diverse pieces of information, competencies that are not easily replaced by AI outputs [7], [8].

Encouraging students to critically evaluate AI-generated content, to question assumptions, and to verify solutions is essential to cultivating deeper learning and professional competence [9]. Integrating AI tools into coursework should thus be accompanied by instructional strategies that advance active learning, reflection, and metacognition [10]. Recent research focus on evaluating AI across Bloom's taxonomy levels [11] aligns with pedagogical goal by highlighting the varying cognitive demands where AI currently excels or struggles, thereby informing targeted educational interventions [12], [13].

A. LITERATURE REVIEW

1) Generative AI (GAI) and Engineering Education

Artificial intelligence (AI), and more recently generative artificial intelligence (GAI), is increasingly shaping research, teaching, and professional practices in engineering education. AI-based systems are becoming embedded in engineering workflows and educational environments, augmenting human cognitive and professional practices through algorithmic systems that assist with tasks such as information retrieval, design exploration, and data analysis. As Johri notes, AI-driven tools extend human cognition by redistributing aspects of reasoning and problem solving across human-machine systems, enabling engineers and learners to manage increasing levels of complexity in technical work [14].

Recent advances in GAI, particularly systems built on foundational models such as large language models (LLMs) have further expanded these capabilities by allowing users to generate explanations, code, design ideas, and analytical responses through natural language interaction. In educational contexts, these systems offer opportunities to support both instructional practices and self-directed learning by providing on-demand explanations, examples, and problem-solving assistance.

At the same time, the integration of GAI into engineering education raises important pedagogical and ethical considerations. Because these systems generate responses based on probabilistic patterns in training data rather than true semantic understanding, their outputs may contain inaccuracies, biases, or misleading information [15]. Consequently, literature emphasizes the importance of cultivating critical AI literacy among engineering students so that they can interpret, evaluate, and responsibly use AI-generated outputs. As the adoption of GAI tools continues to expand, systematic evaluation of their capabilities and limitations within authentic engineering learning contexts becomes

essential. In particular, understanding how LLMs perform on engineering tasks of varying cognitive complexity and how their responses compare with human reasoning can inform responsible pedagogical integration of AI tools in engineering curricula.

2) Evaluations of LLM Performance in STEM Contexts

Recent empirical studies have begun to systematically evaluate the performance of large language models (LLMs) on academic assessments and domain-specific tasks. For example, Kung *et al.* [16] examined ChatGPT's performance on the United States Medical Licensing Examination, while Frieder *et al.* [17] evaluated the mathematical capabilities of ChatGPT (two versions) and GPT-4 using newly created datasets that focus on advanced mathematical reasoning tasks. Within STEM education, emerging research has also explored how LLMs perform on structured scientific and engineering problem-solving tasks. For instance, Wang *et al.* investigated ChatGPT's ability to solve problems from a college-level engineering physics course, including both well-specified textbook-style problems and under-specified real-world problems. Their findings showed that ChatGPT successfully solved approximately 62.5% of well-specified problems but only 8.3% of open-ended problems, highlighting substantial limitations when tasks required modeling real-world situations or making assumptions about missing information [18]. Analysis of incorrect solutions further revealed several failure modes, including difficulties constructing accurate models of physical systems, generating reasonable assumptions when data were incomplete, and performing reliable numerical calculations. These results suggest that while LLMs can effectively retrieve and apply known formulas or concepts in well-structured scenarios, their reasoning performance becomes less reliable in more open-ended problem-solving contexts. A recent work has also examined LLMs as conversational AI tutors in engineering and computer science education [19]. They evaluated several LLMs, including ChatGPT, Claude, Gemini, and Meta AI—on an open-ended computer architecture question using metrics such as accuracy, persuasiveness, and depth of explanation. Their results show that although models often produced largely accurate responses, the quality of explanations varied across systems, highlighting the importance of evaluating LLMs that assess not only correctness but also the quality of AI-generated responses.

Within engineering education specifically, studies have also begun to examine the pedagogical integration of generative AI tools. For example, El Fathi *et al.* [20] explored the use of ChatGPT as a supplemental instructional tool in an introductory thermodynamics course and reported improvements in conceptual understanding within an inquiry-based learning environment. At the same time, scholars have raised concerns regarding the reliability of LLM outputs in structured analytical or high-stakes problem-solving settings [21]. Although these studies demonstrate that LLMs can perform competitively on certain academic benchmarks,

most evaluations rely primarily on aggregate accuracy measures or isolated question sets. Cognitive complexity is often treated implicitly rather than explicitly operationalized. Furthermore, many existing studies focus on a single disciplinary domain, limiting insight into whether LLMs exhibit consistent strengths or weaknesses across engineering fields. These limitations motivate the need for more structured evaluation frameworks that explicitly incorporate cognitive levels and cross-domain comparisons when assessing LLM capabilities in engineering education.

3) Reliability of GAI and the Need for Structured Evaluation

A well-documented limitation of LLMs is their propensity for hallucinations that is, generating plausible sounding but incorrect or fabricated information [22]–[24]. In technical domains such as engineering, hallucinations can lead to erroneous conclusions or unsafe recommendations if not carefully scrutinized. This inherent uncertainty shows the importance of not treating AI outputs as infallible but rather as starting points for critical analysis and validation [25]. In another study, Fulsher, *et al.* [26] reviewed how generative AI can both generate and correct misinformation within educational contexts. Their analysis shows that most research focuses on identifying individual student misconceptions, with less attention to community- and system-level misinformation risks. While hallucination risk and ethical concerns provide important contextual motivation for structured AI evaluation in education, the present study does not empirically quantify misinformation rates. Instead, ethical considerations are addressed conceptually through the instructional guidelines presented in Section IV.

As AI tools become increasingly accessible, their integration into engineering education offers exciting opportunities for enhancing student learning and engagement. However, encouraging responsible and reflective use of AI is critical to ensure that students develop the skills to critically evaluate AI outputs rather than passively accepting them [27], [28]. This study provides a framework to support such engagement by highlighting the varying capabilities and limitations of AI models when applied to engineering problems across multiple cognitive levels.

4) Bloom's Revised Taxonomy as a Evaluation Framework

Bloom's Revised Taxonomy provides a widely used framework for designing and evaluating educational assessments by categorizing cognitive processes into hierarchical levels—remembering, understanding, applying, analyzing, evaluating, and creating [29], [30]. In engineering education, Bloom-aligned learning objectives are frequently used to structure course assessments so that students progressively develop conceptual understanding, analytical reasoning, and problem-solving abilities. As a result, Bloom's taxonomy plays an important role not only in curriculum development but also in the design of assessment tasks that measure different levels of cognitive engagement [31], [32].

Despite its extensive use in educational assessment design, Bloom's taxonomy has only recently begun to be explored as a framework for evaluating the capabilities of LLMs. In LLM research, model performance is commonly assessed using benchmarks, which are standardized datasets of tasks designed to evaluate capabilities such as language understanding, reasoning, or mathematical problem solving. However, benchmark evaluations typically report aggregate accuracy without distinguishing the cognitive processes required by different tasks. Recent work has begun to address this limitation by mapping benchmark tasks to Bloom's taxonomy and analyzing the cognitive levels represented in widely used LLM evaluation datasets [33]. These analyses show that many benchmarks predominantly assess lower and mid-level cognitive skills such as remembering, understanding, and applying, while higher-order reasoning tasks remain underrepresented. Another study evaluated GPT4 on 307 psychosomatic medicine multiple-choice exam questions and reported high overall performance. Using Bloom's taxonomy to qualitatively analyze incorrect responses, they found that errors occurred primarily at the remembering and understanding levels, with additional failures in applying concepts to new situations [34]. In addition, recent work on AI-supported learning environments proposes Bloom-informed prompt design strategies to scaffold higher-order thinking when interacting with LLMs [35]. Together, these studies suggest that Bloom's taxonomy provides a useful cognitive and evaluation framework for analyzing how LLMs perform across different types of tasks.

Building on these literature, the present study applies a Bloom-aligned evaluation framework to engineering education contexts, evaluating LLM performance on questions designed to represent multiple cognitive levels across undergraduate engineering courses.

5) Pedagogical Implications of AI-Supported Learning

Evaluating AI performance on Bloom-aligned tasks can also provide insights into how AI tools may complement human learning processes. For example, comparing model responses under different instructional conditions, such as with and without contextual guidance, can reveal how AI systems respond to varying levels of scaffolding [36]. Such analyses may encourage students to view AI not merely as a source of answers but as a collaborative tool that requires critical interpretation and verification [37].

Exposure to AI-generated reasoning including both correct and flawed responses can also foster metacognitive awareness, encouraging students to question assumptions, cross-check information, and reflect on their problem-solving strategies [9], [38]. Consequently, responsible integration of AI tools into engineering education may support active learning, critical thinking, and ethical technology use [39], [40]. Preparing students for a future where human–AI collaboration is common therefore requires instructional approaches that emphasize both effective AI use and professional judgment [41].

6) Research Gap and Connection to the Present Study

While prior studies demonstrate that LLMs can perform competitively on standardized examinations and domain-specific benchmarks, many evaluations rely on single-discipline assessments or isolated problem sets, with results typically reported using aggregate accuracy metrics. As discussed earlier, benchmark-based evaluations often treat cognitive complexity implicitly, rather than explicitly operationalizing the type of reasoning required by a task. Although recent studies have begun to map benchmark datasets to Bloom's taxonomy, these analyses suggest that existing benchmarks disproportionately assess lower- and mid-level cognitive processes, leaving higher-order reasoning tasks comparatively underrepresented. Furthermore, prior evaluations rarely distinguish between task difficulty and cognitive operation, making it difficult to determine whether reported performance reflects genuine reasoning capabilities or pattern-based answer generation. These methodological limitations highlight the need for structured, cognitively differentiated evaluation frameworks, particularly within authentic engineering education contexts.

To address these gaps, the present study adopts a multi-course, Bloom-aligned evaluation framework to examine LLM performance on questions spanning multiple cognitive levels in undergraduate engineering courses. Unlike prior studies that rely primarily on single-discipline benchmarks or accuracy-only metrics, this approach combines correctness-based evaluation with rubric-guided analysis of reasoning quality, allowing differentiation between surface-level answer selection and deeper explanatory reasoning across cognitive categories. In addition, by examining model responses under different instructional conditions and comparing them with human student performance, this study provides contextual grounding for interpreting LLM capabilities in educational settings.

However, this study shares limitations common in exploratory LLM evaluation research, including a finite instructor-designed item bank and reliance on descriptive statistics. These constraints are acknowledged and frame the findings as methodological illustration rather than definitive validation.

B. RESEARCH PROBLEM AND KNOWLEDGE GAP

Despite growing empirical studies evaluating LLMs in education, existing work primarily reports descriptive accuracy metrics on isolated tasks or standardized benchmarks. Few studies systematically examine LLM performance across multiple engineering domains while explicitly differentiating cognitive task types using a validated learning taxonomy. Furthermore, prior research rarely distinguishes whether observed performance differences reflect task difficulty, prompt design, or underlying cognitive operations.

To address this gap, this study investigates how Bloom's Revised Taxonomy can serve as a structured analytical framework for examining LLM performance across distinct engineering domains.

This work is guided by the following research questions:

RQ1: How do LLMs perform across different cognitive levels defined by Bloom's Revised Taxonomy in undergraduate engineering courses?

RQ2: Do LLMs exhibit consistent strengths or weaknesses across engineering domains when tasks are aligned with cognitive levels?

RQ3: How does the availability of instructional context influence LLM performance across cognitive categories?

RQ4: How does LLM performance compare with human learners under controlled conditions when evaluated using Bloom-aligned tasks?

This study positions its contribution as methodological (a structured Bloom-aligned evaluation framework), empirical (cross-course comparative findings), and conceptual (analysis of LLM reasoning patterns through a cognitive lens).

C. RESEARCH OBJECTIVES

This study evaluates LLMs across four engineering courses: *Thermodynamics*, *Computer Programming*, *Systems and Control* and *Circuits for Engineering Technology*. For each course, we developed 12 multiple-choice questions (MCQs) that span all six levels of Bloom's Revised Taxonomy, from remembering and understanding to analyzing and creating. These questions were designed not to replicate textbook problems, but to assess a model's ability to interpret, reason, and apply knowledge in a pedagogically meaningful way. Importantly, we compare AI performance under two distinct conditions: 1) without any instructional context (zero-shot inference), and 2) with access to curated reference material, simulating the kind of support a student might receive during open-resource learning. We will also compare the LLMs performance with human in *Computer Programming* to evaluate how closely these models align with typical student outcomes and where meaningful gaps in reasoning or accuracy remain.

By applying a structured and comparative approach, this study contributes to the growing body of research on AI in education with three key objectives:

- 1) To measure the accuracy and reasoning quality of LLMs across a range of engineering domains.
- 2) To assess whether LLMs exhibit consistent strengths or weaknesses across different levels of cognitive complexity.
- 3) To evaluate how the presence of instructional material affects AI performance on discipline-specific problem sets.

The two-condition approach, contrasting zero-shot and context-assisted scenarios, also investigates whether providing curated instructional content can reduce the likelihood of hallucinations and improve reasoning fidelity. Understanding these dynamics will help educators and students recognize when and how to appropriately leverage AI tools.

The remainder of this paper is organized as follows. The Methodology section describes the multi-course evaluation

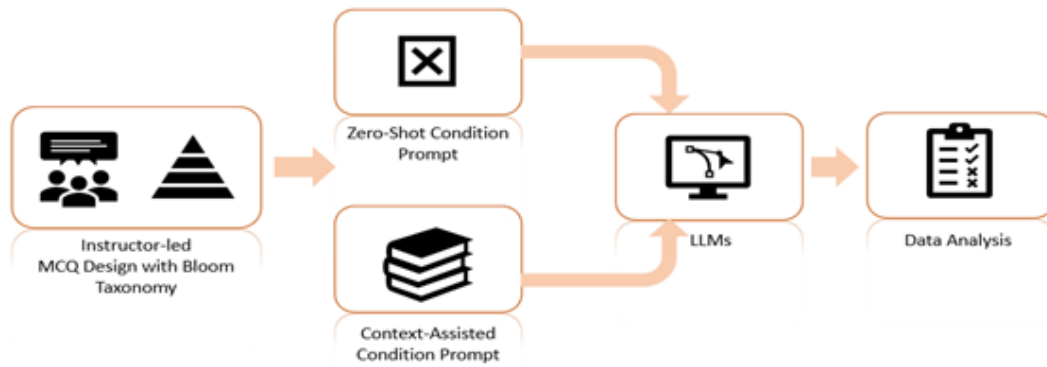


FIGURE 1: Multiple-choice questions were designed by instructors using Bloom's Taxonomy and administered to LLMs under two prompting conditions zero-shot (without content) and context-assisted (with content)

design, the development of rubric-based measures, and the construction of prompt used to assess LLM performance. The Results and Analysis section presents comparative findings across models, cognitive levels, and human-AI performance benchmarks. The Discussion examines the pedagogical implications of these results, highlighting opportunities and cautions for integrating LLMs into engineering education. Finally, the Conclusion summarizes key insights, acknowledges limitations, and identifies directions for future research.

D. NATURE OF CONTRIBUTION

The primary contribution of this study is methodological and exploratory rather than theoretical generalization. Specifically, this work introduces and demonstrates a Bloom-aligned evaluation framework for analyzing large language model responses across engineering domains. While empirical results are presented, they should be interpreted as illustrative applications of this structured assessment approach within the selected task set and participant cohort. The study does not claim to establish validated cognitive equivalence between LLMs and human learners, nor does it propose a formal theory of LLM cognition. Instead, it offers a structured analytical lens for examining AI-generated responses in educational contexts.

II. METHODOLOGY

This study employs a structured, comparative, and cross-sectional design to evaluate the performance of large language models (LLMs) on engineering problem-solving tasks across four undergraduate engineering courses: *Thermodynamics*, *Computer Programming*, *Systems and Control* and *Circuits for Engineering Technology*. The evaluation focuses on two distinct testing conditions:

- 1) Zero-shot inference (without content): LLMs answer questions without access to instructional or reference material.
- 2) Context-assisted inference (with content): LLMs are provided with carefully curated reference content re-

lated to each course before answering.

Each model was evaluated on a common set of 12 multiple-choice questions (MCQs) per course, designed to represent a broad range of cognitive tasks defined by Bloom's Revised Taxonomy. The experimental procedure is summarized in Fig. 1, which illustrates the two testing conditions as well as the evaluation workflow for assessing accuracy and reasoning quality across the four courses.

A. QUESTION DEVELOPMENT

A total of 48 MCQs were developed, 12 for each course, by a team of instructors with domain expertise. These questions were not directly taken from textbooks, but rather designed to evaluate students' understanding, application, and reasoning abilities based on common engineering principles. Each question was categorized into one of the six Bloom's cognitive levels as shown in Table I.

To enhance methodological transparency and replicability, each question was classified into a Bloom cognitive level using explicit operational criteria derived from Krathwohl's revision of Bloom's taxonomy. Rather than relying solely on instructor labeling, classification was guided by the primary cognitive operation required to correctly solve the task.

Questions were reviewed independently by two instructors with domain expertise to ensure alignment between the dominant cognitive operation and assigned Bloom category. Disagreements were resolved through discussion to achieve consensus classification. This operationalization approach reduces reliance on subjective labeling and clarifies how cognitive distinctions were applied in task design.

Although multiple-choice questions are traditionally associated with lower-order assessment, prior assessment research demonstrates that well-designed MCQs can capture higher-order cognitive processes when distractors require evaluation of assumptions, synthesis of relationships, and judgment of alternatives. In this study, Evaluate and Create-level items were structured to require comparative reasoning and justification rather than simple recall.

For example, an Apply-level thermodynamics item required substitution into a known equation under modified boundary conditions, whereas an Analyze-level item required interpretation of variable relationships before determining the appropriate method. An Evaluate-level circuits question required justification of the most appropriate compensation method among alternatives based on system trade-offs.

Questions involved a mix of conceptual reasoning, interpretation of concepts, diagrams, and graphs, short quantitative calculations and design or judgment-based scenarios (in evaluating and creating levels).

Each question included four answer choices, with only one correct answer. The explanations generated by the AI were also collected and evaluated.

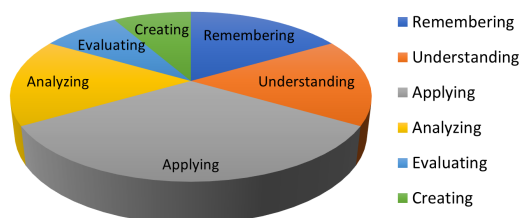


FIGURE 2: Distribution of questions per Bloom's taxonomy level across a course

Fig. 2 shows the distribution of assessment questions across Bloom's taxonomy levels in a course. All the courses followed the same distribution. The majority of questions fall under the Applying category, indicating a strong emphasis on practical problem-solving and the ability to use learned concepts in new contexts. A moderate number of questions assess Understanding and Remembering, reflecting attention to conceptual comprehension and foundational knowledge. In contrast, fewer questions target higher-order skills such as Analyzing, Evaluating, and Creating, suggesting limited but deliberate opportunities for students to engage in critical thinking, synthesis, and innovation. Overall, the distribution demonstrates a well-structured cognitive progression, with a predominant focus on applied learning supported by conceptual understanding.

B. REFERENCE MATERIAL FOR CONTEXT-ASSISTED RUNS

For *Thermodynamics*, a concise reference packet was prepared to simulate instructional material available to students during open-book or guided learning situations. These packets contained key definitions and equations; Sample problem summaries (2-3 worked-out examples); Excerpts from thermodynamic or engineering property tables (where applicable); Process diagrams or schematics

For the *Computer Programming* course, the lecture material relevant to each question was provided to the LLMs.

All content was consolidated into a single file that included: Explanations of the underlying concepts; Simple code demonstrations or examples; Illustrations of common errors associated with the concept; and Flowcharts depicting the logic or process related to the topic.

This material was fed into the model as context prior to question presentation during the context-assisted condition.

C. AI MODELS EVALUATED

Three prominent LLMs were evaluated:

- ChatGPT (GPT-5, OpenAI)
- Claude (Anthropic)
- Grok (xAI)

Each model was tested under both conditions (zero-shot and context-assisted) across all four courses. The same prompts and input structure were used for all models to ensure consistency. All models were accessed through their publicly available interfaces using default system configurations at the time of testing. Decoding parameters such as temperature, top-p, maximum tokens, and random seed were not manually modified. Each question was presented once per condition (zero-shot and context-assisted), and responses were recorded without multiple stochastic runs or aggregation procedures. This approach reflects typical educational usage scenarios rather than controlled model benchmarking experiments.

The following standardized prompting template was used:

You are an expert engineering tutor tasked with answering undergraduate-level questions. Your responses will be evaluated for correctness, reasoning quality, and alignment with Bloom's Revised Taxonomy.

*Please carefully read the question and provide:
Final Answer: The chosen option (A, B, C, or D).*

Step-by-Step Reasoning: A clear explanation of how you reached the answer. Explicitly justify each step (e.g., conceptual recall, application of formula, interpretation of tables/diagrams).

Bloom's Level Justification: Identify which level of Bloom's Revised Taxonomy (Remember, Understand, Apply, Analyze, Evaluate, Create) this question primarily tests, and explain why.

Do not guess randomly. If the question requires a calculation, show intermediate steps. If you are uncertain, state your confidence and reasoning.

Each question was presented individually. The model's output included both the selected answer and its reasoning.

Each model's response was scored on two metrics: Accuracy (Correct/Incorrect); The selected answer was compared to the key. Scores were aggregated across each course, and each Bloom level, each model. Reasoning Quality (1-5 scale): A average rubric-based reasoning score was assigned based on explanation depth, correctness, and clarity given in Table II.

TABLE I: Distribution of questions across Bloom's taxonomy levels

Bloom's Level	Operational Cognitive Definition in This Study	# of Questions (per course)
Remember	Retrieval of factual knowledge without transformation	2
Understand	Explanation or interpretation of conceptual relationships	2
Apply	Execution of known procedures in new but structured contexts	4
Analyze	Decomposition of systems into relationships and cause-effect reasoning	2
Evaluate	Justification of choices based on criteria or trade-offs	1
Create	Generation of novel solution structures or design alternatives	1

TABLE II: Scoring rubric for evaluating reasoning quality in AI-generated responses

Score	Description
5	Clear, concise, and correct explanation demonstrating insight and correct logic
4	Mostly correct logic; minor omissions or simplifications
3	General understanding; lacks specificity or detail
2	Vague or partially incorrect explanation
1	No reasoning, incorrect reasoning, or unrelated content

D. RATIONALE FOR SELECTING ENGINEERING COURSES

The four engineering courses, *Thermodynamics*, *Computer Programming*, *Systems and Control*, and *Circuits* were intentionally selected to represent distinct cognitive demands and domains of engineering learning that align with different levels of Bloom's taxonomy and capture the breadth of challenges encountered in engineering education.

Thermodynamics emphasizes conceptual understanding, multi-step problem solving, and the ability to apply fundamental laws of energy and entropy to real-world systems. Evaluating LLM performance in this domain tests the model's ability to: Interpret word problems that require quantitative reasoning and symbolic manipulation; Translate conceptual understanding into mathematical modeling and stepwise derivation; and explain the physical meaning behind formulas critical for higher-order cognitive skills such as analysis and synthesis. This course represents the traditional analytical rigor of mechanical and aerospace engineering.

Computer Programming course focuses on computational thinking, logic, and syntax-based problem solving. Assessing LLM performance here highlights how effectively a model can: Understand and generate code, debug code and predict outputs; Support procedural knowledge and conditional reasoning (IF-THEN logic, loops, recursion); Demonstrate creativity and decision-making capabilities when multiple solutions exist. This domain tests the LLM's precision, attention to detail, and ability to produce exact, verifiable answers, contrasting with the conceptual reasoning needed in *Thermodynamics*.

Systems and Control integrates mathematics, physics, and computation to design feedback systems and predict behavior over time. It was chosen to evaluate the LLM's ability to: Handle interconnected, dynamic systems that require modeling and stability analysis; Apply both ana-

lytical and design-oriented reasoning, such as interpreting transfer functions or designing controllers; Exhibit systems-level thinking understanding cause-effect relationships and optimization trade-offs.

Lastly, the *Circuits* course was chosen because it offers a rigorous environment for examining the reasoning, accuracy, and adaptability of LLMs in solving structured engineering problems. Evaluating LLM performance in this domain tests the model's ability to: Interpret and apply engineering principles such as Ohm's law, Kirchhoff's laws, and network theorems within both direct and multi-step analytical contexts; Perform quantitative reasoning and formula manipulation accurately while maintaining unit consistency and numerical precision across problem-solving steps; Explain circuit behavior conceptually, demonstrating understanding of current, voltage, resistance, and power relationships rather than relying solely on pattern-based computation and Integrate problem-solving logic with real-world engineering applications, including troubleshooting, design interpretation, and the justification of assumptions in circuit analysis. This domain therefore provides a balanced test of both the computational competence and conceptual reasoning that are central to engineering technology practice.

E. DATA ANALYSIS

Data were analyzed using a combination of descriptive and comparative statistical methods to examine LLM performance across courses and cognitive levels. Descriptive analysis focused on accuracy frequency, percentage of correct responses, and score distribution across cognitive levels, while comparative analyses provided insights into differences across models, courses, and Bloom's taxonomy levels.

The data analysis examined multiple dimensions of LLM performance to provide both quantitative and qualitative insights. Overall accuracy and average reasoning scores were calculated for each course, model, and testing condition to establish baseline performance patterns. To further explore cognitive variability, scores were distributed across Bloom's taxonomy levels, *i.e.* remembering, understanding, applying, analyzing, evaluating, and creating, to identify where each model demonstrated strength or limitation. Comparative averages were then analyzed to reveal course-specific trends, such as higher accuracy in procedural programming tasks compared to conceptual reasoning in thermodynamics. In addition, qualitative coding was applied to classify common error patterns, including misinterpretation of tabular

data, incorrect assumptions, incomplete reasoning chains, and overgeneralized conclusions. This combined approach enabled a comprehensive assessment of how the models performed across domains and cognitive levels.

III. RESULTS AND ANALYSIS

This section presents the comparative performance of three LLMs, GPT-5, Claude, and Grok evaluated across multiple engineering courses and cognitive levels based on Bloom's taxonomy. The analysis examines how each model performed on tasks ranging from foundational recall to higher-order analytical reasoning, using average evaluation scores on a five-point scale. Results are organized by both course context and cognitive demand. Together, these findings provide an illustrative comparison of how the evaluated LLMs responded to Bloom-aligned tasks within the selected engineering contexts. Rather than establishing generalized claims about model cognition, the results demonstrate how a structured taxonomy-based framework can be used to differentiate performance patterns across cognitive categories and domains.

A. OVERALL LLMS PERFORMANCE ACROSS COURSES

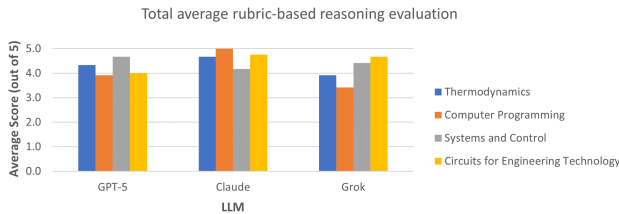


FIGURE 3: Comparison of total average rubric-based reasoning scores (out of 5) achieved by three large language models i.e. GPT-5, Claude, and Grok, across the selected engineering courses

Fig. 3 illustrates the comparative total average rubric-based reasoning scores obtained by the three LLMs, GPT-5, Claude, and Grok, across the courses *Thermodynamics*, *Computer Programming*, and *Systems and Control*. Within the constraints of the selected tasks and rubric criteria, Claude exhibited higher average reasoning scores across the evaluated courses. These results reflect performance patterns specific to the structured assessment design rather than definitive rankings of model capability.

In *Thermodynamics*, both GPT-5 and Claude performed comparably (average scores above 4.0 out of 5), indicating that both models handled conceptual and analytical reasoning effectively. However, in *Computer Programming*, and *Circuits*, Claude outperformed the others, achieving an average near 4.7 and 4.4 out of 5 respectively, reflecting superior precision and logical structuring in the required tasks. GPT-5 maintained stable but lower performance, while Grok lagged significantly, suggesting limited adaptability to

algorithmic or code-generation problems. In *Systems and Control*, GPT-5 slightly outperformed Claude and Grok, indicating stronger integrative reasoning and system-level understanding.

Overall, the results suggest that while all three models demonstrated competency across engineering problem types, Claude's performance was the most consistent, GPT-5 showed greater adaptability to complex system reasoning, and Grok displayed higher variability across contexts. These differences reflect the models' distinct training emphases: Claude's linguistic coherence and GPT-5's structured reasoning capabilities highlighting the importance of aligning LLM selection with course-level cognitive demands.

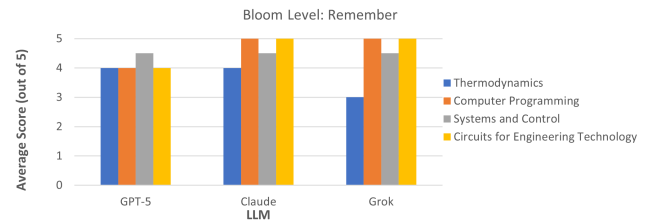


FIGURE 4: Average rubric-based reasoning scores (out of 5) for GPT-5, Claude, and Grok at the Bloom Level: *Remember* across selected engineering courses

Fig. 4 presents the average evaluation scores for the *Remember* level, which included two questions per course designed to assess recall of fundamental concepts, definitions, and formulas. As shown in Fig. 4, all three models performed strongly on recall-based tasks. GPT-5 and Claude achieved comparable scores in *Thermodynamics* and *Systems and Control* (around 4-4.5 out of 5), while Grok scored slightly lower in *Thermodynamics*. In *Computer Programming*, and *Circuits*, Claude and Grok reached perfect scores, reflecting high reliability in factual recall. Overall, LLMs demonstrated strong knowledge retrieval at the *Remember* level, with minimal variation across courses.

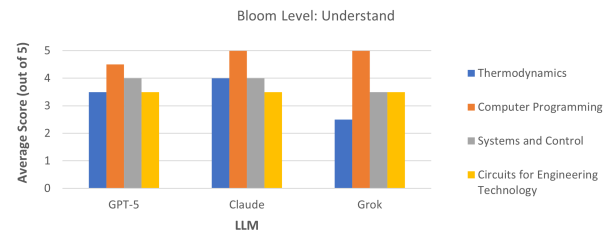


FIGURE 5: Average rubric-based reasoning scores (out of 5) for GPT-5, Claude, and Grok at the Bloom Level: *Understand* across selected engineering courses

As shown in Fig. 5, model performance at the *Understand* level remained strong overall, with higher variation across courses. In *Thermodynamics*, Claude outperformed GPT-5 and Grok, indicating better conceptual interpretation and explanation accuracy. All three models performed excep-

tionally well in *Computer Programming*, reaching or nearing perfect scores, reflecting clear comprehension of code logic and purpose. In *Systems and Control*, GPT-5 and Claude achieved comparable scores around 4.0, while Grok trailed slightly. In *Circuits*, all three models performed in a moderate range. Overall, Claude demonstrated the most consistent understanding across courses.

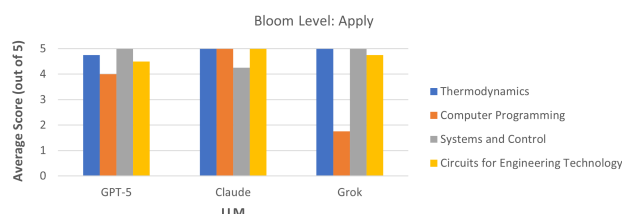


FIGURE 6: Average rubric-based reasoning scores (out of 5) for GPT-5, Claude, and Grok at the Bloom Level: *Apply* across selected engineering courses. Each course included four application-based questions. GPT-5 and Claude maintained strong application performance, while Grok's results varied by domain

Fig. 6 shows the LLMs performance based on four application-focused questions per course. All models demonstrated strong application skills, though with notable differences across courses. In *Thermodynamics* and *Circuits*, all three models achieved similarly high scores (around 4.8-5.0), indicating solid ability to apply principles to problem-solving scenarios. In *Computer Programming*, Claude achieved the highest score (5.0), while GPT-5 performed moderately well and Grok scored significantly lower, suggesting challenges with code implementation or contextual application. In *Systems and Control*, GPT-5 and Grok both reached perfect scores, highlighting strong proficiency in applying mathematical and systems-based reasoning. Overall, GPT-5 and Claude displayed consistent strength at the *Apply* level, with Grok showing greater variability across domains.

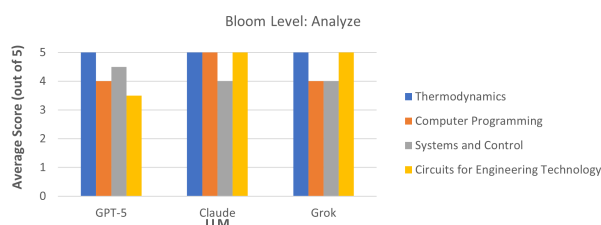


FIGURE 7: Average rubric-based reasoning scores (out of 5) for GPT-5, Claude, and Grok at the Bloom Level: *Analyze* across selected engineering courses

As illustrated in Fig. 7, performance at the *Analyze* level, based on two analysis-oriented questions per course, remained high across models, with limited variation. In *Thermodynamics*, all three models achieved perfect scores,

reflecting strong analytical reasoning and accurate interpretation of system relationships. In *Computer Programming*, Claude slightly outperformed GPT-5 and Grok, indicating stronger skill in breaking down algorithmic structures or debugging logic. In *Systems and Control*, GPT-5 achieved the highest average (around 4.5), while Claude and Grok followed closely. In *Circuits*, Claude and Grok got perfect scores while GPT5 achieved the lowest (3.5). Overall, all models demonstrated effective analytical reasoning, with Claude showing consistency across domains and GPT-5 excelling in system-level interpretation.

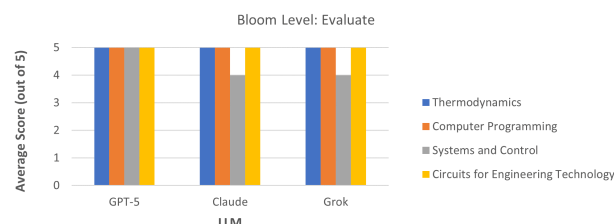


FIGURE 8: Average rubric-based reasoning scores (out of 5) for GPT-5, Claude, and Grok at the Bloom Level: *Evaluate* across selected engineering courses

On *Analyze* level as shown in Fig. 8, GPT-5 and Claude both achieved the maximum evaluation score (5/5) across all courses, reflecting strong consistency and accuracy in judgment-based and critical analysis tasks. Grok displayed comparable performance in most courses but showed noticeably lower scores in *Systems and Control*, suggesting reduced reliability when dealing with control-oriented analytical reasoning. Overall, the results indicate that GPT-5 and Claude demonstrate robust higher-order evaluative capabilities across diverse engineering contexts, while Grok's performance appears more domain-dependent.

Fig. 9 illustrates how each large language model (LLM) performed on higher-order creative reasoning tasks requiring synthesis, innovation, or open-ended problem framing. Claude consistently achieved the highest or near-maximum scores across all four courses, indicating strong generative and design-oriented reasoning. GPT-5 demonstrated moderate creative ability, performing best in *Systems and Control* but showing lower synthesis depth in *Thermodynamics* and

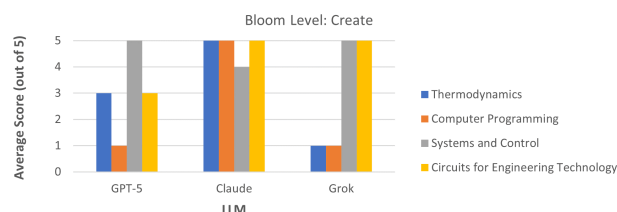


FIGURE 9: Average rubric-based reasoning scores (out of 5) for GPT-5, Claude, and Grok at the Bloom Level: *Create* across selected engineering courses

Computer Programming. Grok exhibited the weakest creative performance overall, particularly in conceptual design and problem-generation tasks. These results suggest that while Claude demonstrates robust cross-domain creative reasoning, GPT-5's generative quality varies by context, and Grok's creative outputs remain limited in abstraction and originality.

B. PERFORMANCE ACROSS BLOOM'S LEVELS

As shown in Fig. 10, the performance of GPT-5, Claude, and Grok varied across the six Bloom's taxonomy levels, *Remember*, *Understand*, *Apply*, *Analyze*, *Evaluate*, and *Create*. Overall, Claude achieved the highest and most consistent scores, maintaining averages above 4.0 across all levels and reaching perfect performance (5.0) at the *Apply*, *Evaluate*, and *Create* stages. GPT-5 followed closely, performing strongly through *Apply* and *Analyze* levels but showing a noticeable drop at the *Create* level, indicating some limitations in open-ended or innovative reasoning tasks. Grok demonstrated greater variability, performing comparably at lower levels but showing declines in *Understand* and *Create*, suggesting challenges with conceptual transfer and generative synthesis. These patterns suggest that, within this Bloom-aligned framework, differences emerge in how models respond to tasks categorized as higher cognitive levels. However, given the limited item set and task structure, these findings should be interpreted as exploratory rather than as definitive evidence of sustained higher-order reasoning capability.

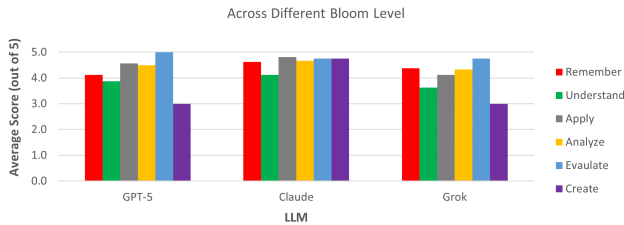


FIGURE 10: Average rubric-based reasoning scores (out of 5) for GPT-5, Claude, and Grok across Bloom's levels (*Remember*, *Understand*, *Apply*, *Analyze*, *Evaluate*, and *Create*). Claude showed the highest overall consistency, while GPT-5 maintained strong analytical and application-level performance

C. COMPOSITE PERFORMANCE INDEX (CPI)

To evaluate and compare the overall performance of the three LLMs, a Composite Performance Index (CPI) was developed. The CPI provides a single, normalized measure of each model's effectiveness based on four key dimensions of response quality: Accuracy, Completeness, Efficiency, and Support. Higher CPI scores correspond to better-performing LLM outputs across the evaluated criteria.

Accuracy (A) measures whether the model produced the correct final answer on the first attempt. It was assigned a

binary value: 1 if the initial response was correct, and 0 if incorrect.

Completeness (C) reflects the degree to which the response contained all necessary reasoning steps, explanations, and contextual detail. It was rated using a rubric score (1-5) and normalized by dividing the score by 5 to yield a value between 0 and 1.

Efficiency (E) captures how quickly and concisely the model reached a complete answer. This was quantified using the number of follow-up prompts required to achieve a correct and complete response, expressed as given in Eq. (1):

$$E = \frac{1}{1 + N_f} \quad (1)$$

where N_f is the number of follow-up questions. Thus, a model that produced a complete answer without follow-ups received an efficiency score of 1. The efficiency metric reflects conversational compactness (number of follow-up prompts) rather than pedagogical quality and should not be interpreted as implying that fewer interactions are inherently more educationally effective.

Support (S) represents the model's ability to enrich its response through the use of visual examples, contextual expansion, or additional explanatory depth. A value of 1 was assigned if such support was present and 0 otherwise.

Each dimension was normalized to a 0 – 1 scale, and a weighted linear combination was used to calculate the CPI for each model and question as shown in Eq. (2):

$$CPI = 0.3A + 0.3C + 0.2E + 0.2S \quad (2)$$

The weights were selected to emphasize accuracy and completeness (60% combined contribution) as core indicators of conceptual and computational correctness, while efficiency and support (40%) capture the response quality and pedagogical richness of the interaction.

The CPI was calculated for all 12 questions across different Bloom's taxonomy levels in engineering courses, providing a balanced metric to compare the performance of each LLM across diverse cognitive tasks. Higher CPI values indicate greater overall effectiveness, *i.e.*, accurate, complete, efficient, and contextually supportive responses, while lower values reveal areas where the model's performance required more guidance or clarification.

Fig. 11 presents the distribution of the CPI scores for GPT-5, Claude, and Grok for *Thermodynamics* across all 12 questions. Each box represents the variability in model performance, illustrating the median, interquartile range (IQR), and outliers for the aggregated CPI values. The boxplot highlights both the central tendency and consistency of each model's responses.

Claude achieved the highest median CPI, indicating the most accurate and complete first-attempt responses on average. However, its relatively wider interquartile range suggests occasional variability in performance, likely due to fluctuations in the level of reasoning detail provided across

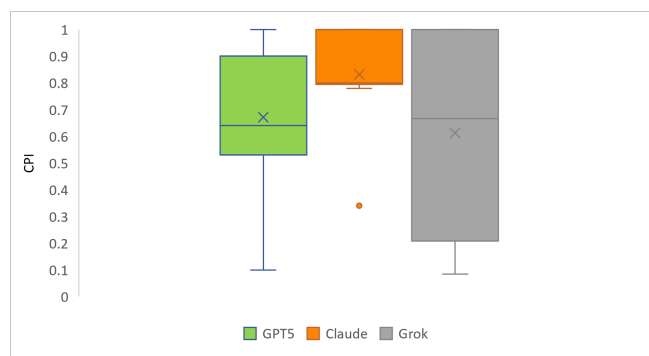


FIGURE 11: Boxplot showing the distribution of Composite Performance Index (CPI) scores for GPT-5, Claude, and Grok across all 12 questions in *Thermodynamics*. Each box represents the interquartile range (IQR), with the central line indicating the median and whiskers showing score variability

question types. GPT-5 demonstrated a slightly lower median CPI but a narrower box, reflecting greater consistency and stability across different Bloom-level questions. Grok exhibited the widest spread and the lowest median, suggesting that while it occasionally produced high-quality responses, it was more prone to incomplete or less efficient answers that required additional follow-ups.

Overall, the boxplot analysis reveals that all three models were capable of generating correct answers but differed in depth, efficiency, and contextual support. Claude tended to excel in high-completeness scenarios, GPT-5 maintained more uniform performance, and Grok showed less reliability across cognitive levels. This visualization provides a concise representation of comparative model performance, emphasizing not only average effectiveness but also the consistency and predictability of each LLM's output quality.

Building on this overview of model performance across cognitive levels, we next examine how access to instructional context influences LLM accuracy, focusing specifically on results from the *Computer Programming* course. Comparing results generated with and without domain-specific content allows us to assess the extent to which these models rely on external information versus internalized reasoning.

D. LLM PERFORMANCE WITH AND WITHOUT CONTENT

The bar chart shown in Fig. 12 compares the total accuracy scores (0-1 scale) achieved by the three LLMs for *Computer Programming* under two different conditions: with content uploaded and without content uploaded. This figure presents the total accuracy of each LLM, scored dichotomously (1 = correct, 0 = incorrect) to isolate performance on factual correctness. Claude achieved perfect accuracy in both conditions, demonstrating stable reliability independent of contextual prompts. GPT-5 also reached perfect accuracy when domain content was supplied and maintained a slightly

lower but still high score without it, indicating mild dependence on contextual cues. Grok showed noticeably lower accuracy overall, performing equally in both conditions but with reduced precision relative to the other models. These results suggest that access to explicit course content modestly benefits GPT-5, has little effect on Claude, and does not substantially change Grok's performance, highlighting differences in intrinsic reasoning robustness among LLM architectures.

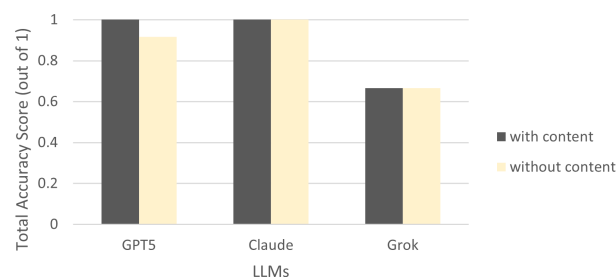


FIGURE 12: Comparison of total accuracy scores (0-1 scale) for GPT-5, Claude, and Grok under two conditions: Zero-shot (without content uploaded) and Context-assisted (with content) for *Computer Programming*

Fig. 13 presents the average rubric-based reasoning scores (0-5 scale) for the three LLMs' explanation quality in *Computer Programming* under two conditions: with supporting content uploaded and without supporting content uploaded. These scores capture each model's ability to clearly, concisely, and correctly justify its answer choice and describe the underlying conceptual reasoning. Claude achieved a near-perfect score without content, and its performance decreased slightly when content was provided, suggesting that its explanations were already highly clear and concise and that additional context did not improve its reasoning. In contrast, GPT-5 produced stronger and more detailed conceptual explanations when content was available, whereas its explanations were comparatively simplified without content. Similar to the accuracy results, Grok exhibited notably weaker explanatory capability in both conditions; however, its performance was marginally higher when no content was uploaded. Overall, these results indicate that Claude and Grok generated better explanations without additional content, whereas GPT-5 benefited substantially from having the supporting material. When comparing the three LLMs, Claude consistently provided the most coherent and conceptually grounded reasoning across computing-related questions.

E. COMPARISON OF LLM AND HUMAN PERFORMANCE

To provide a broader perspective on model effectiveness, the performance of the LLMs on the 12 MCQs was compared with the performance of a cohort of undergraduate students who completed the same set of questions. To examine how students approached and reasoned through a set of 12

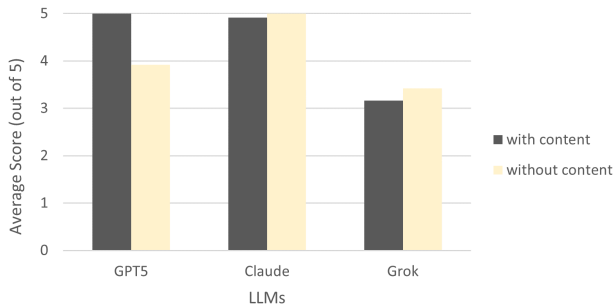


FIGURE 13: Comparison of average reasoning scores (0-5 scale) for GPT-5, Claude, and Grok under two conditions: Zero-shot (without content uploaded) and Context-assisted (with content) for *Computer Programming*

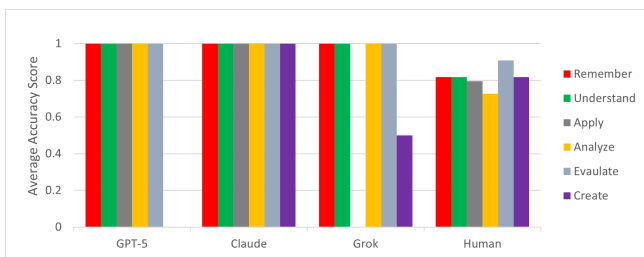


FIGURE 14: Comparison of average accuracy scores (0-1 scale) across Bloom's taxonomy levels for GPT-5, Claude, Grok, and human participants who took the *Computer Programming* course.

MCQs, we conducted an Institutional Review Board (IRB) approved study.

Recruitment emails were sent to undergraduate students who had previously completed the *Computer Programming* course in earlier semesters. A total of 11 students participated in the study. Each participant attended a one-hour proctored session during which they completed the MCQs and provided written explanations for their answer choices. The proctoring ensured academic integrity and confirmed that participants did not consult LLMs or other external resources. Prior to beginning the session, each student provided informed consent. For each student, we then computed an accuracy score representing the number of questions answered correctly out of the 12 total MCQs.

Fig. 14 illustrates the comparative performance of the three LLMs and the human participant cohort on 12 *Computer Programming* MCQs across the six Bloom's cognitive categories, based on the proportion of correct responses. For this comparison, we used the LLMs' scores without any supporting content, ensuring that both LLMs and humans were evaluated solely on their inherent knowledge. Claude demonstrated perfect accuracy (1.0) across all Bloom's levels, indicating a consistently strong grasp of both lower-order and higher-order cognitive tasks. GPT-5 closely followed, achieving near-perfect accuracy in every category

except for a slight reduction in the *Create* category. Grok performed well on *Remember* and *Understand* tasks and also showed competence in higher-order categories such as *Analyze* and *Evaluate*; however, its performance dropped noticeably for *Apply* and *Create*. Human participants showed comparatively lower accuracy overall and exhibited modest variation across categories, with the lowest performance observed in the *Analyze* category.

Within this controlled comparison and limited participant cohort, Claude and GPT-5 achieved higher accuracy on the selected MCQs than the participating students. These findings are context-specific and should not be interpreted as broad claims regarding human-AI comparative cognitive ability.

IV. DISCUSSION

The present study evaluates LLM performance and reasoning patterns on Bloom-aligned MCQ tasks across multiple engineering domains. It does not measure student learning outcomes or experimentally test instructional interventions. Rather, it analyzes how observed response characteristics may inform the design of future LLM-mediated educational activities.

The results illustrate differentiated response patterns among GPT-5, Claude, and Grok when evaluated through a Bloom-aligned assessment structure. Rather than establishing inherent cognitive hierarchies among models, the findings demonstrate how taxonomy-based analysis can reveal variations in explanation depth, procedural execution, and response structure across engineering domains.

Thermodynamics: Quantitative differences also highlighted model-specific reasoning strategies. In a thermodynamics problem involving temperature-dependent specific heats, Claude correctly reasoned that the difference in values led to only a 0.8% change in energy, accurately noting that this deviation is negligible for small temperature intervals. GPT-5 reached a comparable conclusion but with a 5% numerical error attributable to reference-table rounding, while Grok misapplied constants entirely. Notably, GPT-5 was the only model that cited a credible reference, whereas Claude and Grok refused to identify data sources, indicating limitations in source transparency that affect reproducibility and trustworthiness.

Performance at higher cognitive levels revealed further distinctions in reasoning coherence and visualization accuracy. In a numerical multiple-choice question requiring use of thermodynamic tables, all models selected the correct answer; however, when prompted to generate a follow-up P-V diagram, GPT-5 produced a smooth, physically accurate plot, whereas Claude and Grok generated irregular "chunky" curves due to inconsistent sampling of data points. Nonetheless, both Claude and Grok provided rich qualitative insights, describing the hyperbolic relationship between pressure and volume, the physical meaning of compression work, and the energy represented by the shaded area under the curve. On an *Evaluate*-level question that

required justifying the solution through a P-v diagram, only Claude generated an accurate and interpretable figure. GPT-5 and Grok both failed to depict the correct trend, and their second attempts further degraded, emphasizing the need for iterative prompt design and model-specific feedback mechanisms in visual reasoning tasks.

Collectively, these findings suggest that while all models can retrieve and apply core engineering principles, Claude exhibits superior conceptual reasoning and explanation, GPT-5 demonstrates analytical precision and credible reference use, and Grok remains limited by overgeneralization and inconsistent quantitative handling. Pedagogically, this differentiation is critical: From an instructional design perspective, models that provide more elaborated reasoning may offer opportunities for structured critique and reflection activities. However, the educational impact of such use would require empirical validation beyond the scope of the present study.

Computer Programming: As this course primarily cultivates computational thinking, logical reasoning, and syntax-based problem solving, the multiple-choice question format serves as an effective medium for evaluating how well the three LLMs exhibit human-like reasoning by identifying correct answers through the systematic testing of provided options. For *Computer Programming* questions, Claude outperformed both GPT-5 and Grok in terms of accuracy and reasoning. It correctly identified all answers and provided clear, concise, and logically sound explanations that reflected a strong conceptual understanding. This performance was consistent across both methods, *i.e.* with and without content. Moreover, its responses effectively articulated the underlying principles, helping novice programmers connect conceptual knowledge with the given options in the multiple-choice questions.

We found that GPT-5 was generally accurate even when the related content was not provided, and its accuracy improved further when the content was available. In terms of reasoning, it offered clear and logically sound explanations demonstrating strong conceptual understanding for approximately 50% of the questions. For the remaining half, although GPT-5 displayed a reasonable grasp of the underlying logic behind its chosen answers, its explanations often lack specificity or depth. This limitation could pose challenges for novice programmers, as concise but conceptually underdeveloped responses may hinder their ability to fully grasp the underlying principles. Compared to both other LLMs, Grok exhibited lower accuracy across both methods. One of the primary reasons for this underperformance was its inability to interpret code from images, a task that the other models handled effectively. When Grok responded accurately, its explanations were clear and detailed. However, in cases where it failed to identify the correct answer, its behavior was inconsistent; for example, in one instance, it provided two possible answers and asked which one should be preferred. Such inconsistency in responses could be particularly challenging for novice

programmers, as it may introduce confusion or uncertainty rather than reinforcing correct conceptual understanding.

Systems and Control: A key factor influencing performance is the accuracy of the LLMs' internal numerical solvers, or their ability to invoke external ones, when handling quantitative analysis. This is evident in several *Systems and Control* questions that require finding the roots of a system's characteristic equation. The performance of the three LLMs varies significantly, even for equations containing only lower-order polynomial terms. GPT-5 delivers the most consistent and accurate results, with its answers matching numerical benchmarks from MATLAB. In contrast, Grok and Claude struggle to identify the correct roots. This is particularly alarming because without identifying the right roots (or poles of the transfer functions), Grok and Claude are unable to make the right choices for the questions concerning the basic system characteristics. In providing justifications, GPT-5 and Grok demonstrate high confidence in their initial choices and do not alter their answers. Claude, however, frequently proposes an answer initially but, while explaining its rationale, often recognizes errors and changes its selection. This inconsistent behavior could cause significant confusion for students who might rely on Claude as a tutorial tool.

Circuits for Engineering Technology: All three models showed strong understanding of circuit fundamentals and quantitative reasoning. GPT-5 provided concise and accurate answers but often with minimal explanation. Claude and Grok offered fuller reasoning, including verification steps and conceptual discussion, which strengthened their instructional value. Performance differences appeared mainly in procedural and analytical items such as nodal analysis and power-factor correction. GPT-5 initially selected the wrong capacitor value in Question 12, while Claude misidentified the current ratio in the three-phase problem. Grok was accurate throughout but sometimes overanalyzed. Overall, all three models solved the circuit problems effectively. GPT-5 proved efficient for rapid numerical verification, while Claude and Grok provided greater pedagogical depth through detailed explanations and stronger conceptual clarity.

These observations reinforce that generative AI tools must be aligned with learning outcomes and Bloom levels. For lower-order objectives, they can effectively scaffold recall and comprehension. For higher-order objectives, *i.e.* application, analysis, and evaluation, instructors should design guided prompts that require reasoning steps, annotated figures, and explicit justification to prevent superficial correctness. Future work should quantify the impact of such structured prompting on students' critical thinking and concept retention, expanding analysis across larger item sets and diverse engineering domains.

The pedagogical characterizations proposed here (*e.g.*, reflective tutor, accuracy checker) should be understood as design implications inferred from observed response patterns rather than experimentally validated instructional roles.

TABLE III: Guidelines for Integrating LLMs into Engineering Education

Application Context	Purpose / Use Case	Recommended Student Approach	Instructor Guidance and Pedagogical Considerations	Key Cautions
Concept Review and Knowledge Recall	Retrieve definitions, formulas, and physical principles at <i>Remember</i> and <i>Understand</i> levels.	- Use LLMs to clarify concepts or unfamiliar terminology. - Cross-check outputs with course notes or textbooks. - Keep a personal log of corrected AI mistakes.	- Encourage students to justify whether LLM explanations align with accepted physical principles. - Incorporate verification tasks to promote critical evaluation.	Overreliance on factual recall can mask gaps in conceptual reasoning.
Problem-Solving and Equation Application	Work through applied or numerical problems (<i>Apply</i> level).	- Use LLMs to outline solution steps, not just final answers. - Manually verify calculations and assumptions. - Identify and explain each formula used.	- Design assignments requiring students to compare their manual and AI-generated solutions. - Reward process accuracy more than final numeric results.	LLMs may misapply equations or omit boundary conditions; stress validation.
Analytical Reasoning and Systems Thinking	Explore relationships among variables and evaluate design trade-offs (<i>Analyze</i> level).	- Ask LLMs to explain why different outcomes occur. - Diagram reasoning steps to visualize links. - Use structured prompts demanding "explain and justify" reasoning.	Incorporate reflective comparisons between student and AI logic.	LLMs may produce logically coherent but physically invalid explanations.
Critical Evaluation and Decision-Making	Assess alternative solutions or methods (<i>Evaluate</i> level).	- Prompt LLMs to provide pros and cons for each approach. - Critically rank options and justify final choice.	- Use LLM outputs as case studies for ethical or design discussions. - Highlight bias and uncertainty in AI reasoning.	AI confidence $\neq$ correctness; students should check source credibility.
Creative Design and Innovation	Brainstorm ideas for system improvement or engineering solutions (<i>Create</i> level).	- Use LLMs for idea generation and early-stage prototyping. - Annotate AI-generated concepts with own refinements.	- Frame AI as a creative collaborator, not a substitute. - Integrate design critiques comparing AI and student solutions.	AI creativity may reflect pattern imitation rather than innovation.
Code Debugging and Algorithm Development	Analyze logic and syntax errors in computer programs.	- Request LLM explanations for debugging rationale. - Test all AI-generated code in sandbox environments.	- Encourage prompts demanding error analysis, not correction alone. - Require documentation explaining AI-suggested edits.	Risk of unsafe or inefficient code; emphasize verification via testing.
Ethical and Professional Reflection	Evaluate implications of AI use in engineering contexts.	- Reflect on how AI reliance shapes professional responsibility. - Discuss biases or misinformation in outputs.	Integrate reflective writing linking AI use to ethics and safety.	Failure to address ethical use undermines professional competence.

Future research could empirically test these roles through controlled classroom studies comparing learning outcomes under structured LLM-assisted conditions (*e.g.*, reflection-oriented scaffolding versus accuracy verification support), using pre-post assessments or randomized instructional assignments to evaluate impact on student performance and conceptual understanding.

Table III provides a structured guide for effectively integrating LLM into engineering education through purposeful, cognitively aligned use. It outlines how students and instructors can employ AI tools to reinforce conceptual understanding, analytical reasoning, and creative problem-solving across Bloom's levels while maintaining academic rigor and ethical awareness.

The framework positions LLMs as interactive learning partners rather than answer generators, encouraging critical engagement, reflection, and validation of AI-generated reasoning. For instructors, it emphasizes designing prompts and assessments that require justification, comparison, and metacognitive reflection; for students, it highlights strategies to question, verify, and refine AI outputs. Collectively, these practices foster active learning, strengthen professional judgment, and promote responsible human-AI collaboration in engineering contexts.

While the findings provide structured insight into model strengths and limitations across cognitive levels, the translation of these patterns into innovative instructional designs remains to be empirically evaluated. Future classroom-based studies examining learning gains, engagement, and conceptual transfer under LLM-supported conditions are necessary to determine the educational impact of such applications.

V. CONCLUSION

This study explored how large language models respond to engineering assessment tasks when analyzed through Bloom's Revised Taxonomy. Within the constraints of the selected courses, item set, and rubric framework, the models demonstrated differentiated performance patterns across cognitive categories. These results should be interpreted as illustrative applications of a structured evaluation approach rather than as generalized claims of expert-level reasoning.

For engineering instructors, these findings underscore the need to shift from perceiving AI as a replacement for human cognition to treating it as a cognitive amplifier, a partner that can model reasoning processes, provide immediate formative feedback, and promote metacognitive engagement when used strategically. Instructors should design activities that require students to interrogate AI explanations, identify assumptions, and justify alternative reasoning paths. Embedding LLM-assisted exercises into assignments or lab reports can cultivate critical thinking, while structured reflection prompts (*e.g.*, Where might the model's reasoning fail?) reinforce deeper conceptual understanding. Moreover, LLM output can serve as a benchmark for diagnostic feedback, enabling educators to identify common student misconceptions

and to tailor instruction toward conceptual clarity rather than rote performance.

For students, this work highlights both the potential and the responsibility that accompany AI use in technical learning. When leveraged as an interactive learning companion rather than an answer-generator, LLMs can accelerate comprehension, offer diverse solution pathways, and support autonomous exploration. Yet over-reliance on AI risks eroding the very analytical rigor that defines engineering expertise. Students must therefore approach these tools with skeptical curiosity, validating AI-generated solutions, cross-checking equations, and reflecting on the physical or computational meaning behind every result. Developing such habits will be essential for engineers operating in a future where human-AI collaboration defines professional practice.

The findings suggest that Bloom-aligned frameworks may provide a structured lens for analyzing AI-generated responses in educational contexts. Future research should replicate this approach across larger datasets, alternative assessment formats, and diverse institutions to evaluate the stability and transferability of these observations.

The present work should be understood as exploratory and methodological in nature. Its primary contribution lies in demonstrating how a taxonomy-aligned analytical framework can be applied to interpret LLM responses across engineering domains. Broader generalization regarding AI cognition or instructional impact would require larger samples, expanded task sets, and multi-institutional replication.

A. LIMITATIONS AND FUTURE WORK

While the present study offers a detailed comparative analysis of LLM performance and human cognition in engineering education, several limitations should be acknowledged. First, the evaluation relied on a finite set of questions and rubric-based scoring that, while structured and validated, cannot fully capture the breadth of engineering problem-solving encountered in authentic, open-ended design contexts. The analysis also focused on text-based interactions, excluding multimodal capabilities such as diagram interpretation or code generation that could further influence performance outcomes. Additionally, the human cohort represented a limited sample size from a single course context, and differences in prior knowledge, motivation, or test-taking behavior may have influenced the comparative results.

Higher-order cognitive levels (*Evaluate* and *Create*) were represented by a single item per course. These items differed in structure from lower-level questions in that they were designed as integrative, project-oriented scenarios requiring synthesis, justification, or design reasoning rather than discrete procedural execution. As such, their solution processes were more intensive in terms of reasoning depth and assessment criteria. The reduced number of items in these categories reflects this qualitative difference in task structure rather than an omission of higher-order cognition. Nevertheless, the limited item count constrains the statis-

tical robustness of level-specific comparisons. Accordingly, interpretations of Evaluate and Create performance should be considered exploratory. Future research incorporating larger pools of higher-order tasks would enable more stable quantitative comparisons while preserving the complexity of such assessments.

The assessment items used in this study were instructor-designed to support exploratory evaluation within a Bloom-aligned analytical framework rather than to function as a formally validated psychometric instrument. Although Bloom-level assignments were reviewed by domain instructors for alignment with operational cognitive definitions, formal inter-rater reliability statistics (e.g., Cohen's K) were not computed. Additionally, item-level psychometric properties such as difficulty indices and discrimination coefficients were not calculated. Given the limited number of items per cognitive level and the small human cohort, such estimates would not provide stable or generalizable metrics. Future studies employing larger item pools and broader participant samples could incorporate full psychometric validation procedures.

The human benchmark included a single Computer Programming cohort ($N = 11$) and was not randomly sampled. As such, comparisons between LLMs and human participants are limited to this specific course context and MCQ set. The results should therefore be interpreted as contextual rather than representative of broader student populations or generalized human performance across engineering domains. The human comparison serves as an illustrative reference point within the study's exploratory framework rather than as a statistically generalizable benchmark.

Future work should expand this framework to include larger and more diverse datasets, incorporating multi-institutional samples and a broader range of engineering domains. The integration of multimodal LLMs (e.g., models that process images, schematics, and simulations) presents a promising avenue for assessing AI's role in visual reasoning and systems design. Longitudinal studies are also needed to measure how sustained AI exposure impacts students' conceptual retention, metacognitive skills, and ethical decision-making. Finally, developing AI-literate instructional models, where students learn to critique, calibrate, and co-create with AI systems, will be essential for transforming these technologies from assessment novelties into enduring instruments of engineering insight and innovation.

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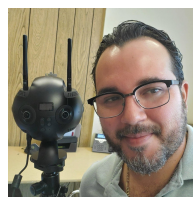
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